

ULIP Integration Document

Version No.: 1.0

Date: 17/09/2024

Project Name: ULIP

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Revision History

Version	Date	Author	Reviewer	Significant Changes
1.0	17/09/2024	Ayush Arya	Nayab Ali	Baseline Version

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Introduction

1.1 Purpose

The purpose of this document is to define and provide details for accessing API for MOPNG system information (information about petrol pumps and natural gas stations on the basis stcode or latitude, longitude and radius) from ULIP system

This document details the following points:

1. **Data Integration between MOPNG and ULIP web service:** Identification of operational points where data integration between **MOPNG** and ULIP web service will be needed.
2. **Technical Approach** for performing Data Integration

The Overall Description

Users will need to share their stcode, or geolocation (latitude, longitude and radius) to access the list of public petrol pumps through the rest API.

Here are some points to keep in mind when accessing MOPNG API data information:

- The API will only share data that is based on the user's provided information.
- ULIP will expose API for getting data from MOPNG API.
- The MOPNG/01 API will take stcode, or geolocation (latitude, longitude and radius) from user and get data.
- radius value should be less than or equal to 100.

1.2 Data Integration between MOPNG System and ULIP application

The ULIP system will receive information from the user and provide data in return.

1.3 MOPNG/01

This API is designed to retrieve information about petrol pumps and natural gas stations. It can be used by providing either the stcode, or geolocation (latitude, longitude and radius) of the desired location.

Please note that when making a request, **all of the JSON keys are mandatory in the payload**. If you choose to use the stcode to access MOPNG API data, then latitude, longitude and radius fields should be left empty.

```
{
  "stcode": "28",
  "latitude": "",
  "longitude": "",
  "radius": ""
}
```

When sending geolocation(latitude, longitude and radius) data, if the user provides geocoordinate latitude, longitude and radius, then the stcode will be empty.

```
{
  "stcode": "",
  "latitude": "14.3473952000",
  "longitude": "78.0001640500",
  "radius": "1"
}
```

	Field Name	Field Description	Format	Required/Not – Required	Length
1	stcode	State Code	[0-9]{0,6}	Required	0-6
2	latitude	Latitude	^[0-9]{2}\.{0,1}[0-9]{2,10}[[0-9]{0,2}]\$	Required	0-13
3	longitude	Longitude	^[0-9]{2}\.{0,1}[0-9]{2,10}[[0-9]{0,2}]\$	Required	0-13
4	radius	Radius	^\d{0,3}(\.\d{1,10})?\$	Required	0-3

1.3.1 Technical Approach

1. All the data will be shared through REST web services.
2. The user will share the input for which information is required.

3. MOPNG API will share details of MOPNG API station information that exists in the system in JSON format.
4. The ULIP system will provide the data in JSON Object format in the response body.

Example:

Request:

The request must be of the following format, where **stcode**, or **geolocation (latitude, longitude and radius)** are passed within the URL post request body in JSON format.

For Staging -

<https://www.ulipstaging.dpiit.gov.in/ulip/v1.0.0/MOPNG/01>

```
{
  "stcode": "",
  "latitude": "14.3473952000",
  "longitude": "78.0001640500",
  "radius": "1"
}
```

Curl URL

```
curl --location 'https://www.ulipstaging.dpiit.gov.in/ulip/v1.0.0/MOPNG/01' \
--header 'Authorization: Bearer eyJhbGciOiJIUzUxMiJ9.eyJzdWIiOiJ1bGlwliwiaWF0IjoxNzI2NTU1ODE0LCJhcHBzIjoZGF0YXB1c2gifQ.a5dxkDgb_kbRXQhm5XfVH5gfj0b_PfXC0DTFWbjsiya7sPbLQsN5sR4-55Tm3hJy4mqSyn9P0HTJqXsdJX5uA' \
--header 'Content-Type: application/json' \
--header 'Accept: application/json' \
--data '{
  "stcode": "",
  "latitude": "14.3473952000",
  "longitude": "78.0001640500",
  "radius": "1"
}'
```

For Production -

<https://www.ulip.dpiit.gov.in/ulip/v1.0.0/MOPNG/01>

```
{
  "stcode": "",
  "latitude": "14.3473952000",
  "longitude": "78.0001640500",
  "radius": "1"
}
```

Curl URL

```
curl --location 'https://www.ulip.dpiit.gov.in/ulip/v1.0.0/MOPNG/01' \
--header 'Authorization: Bearer
eyJhbGciOiJIUzUxMiJ9.eyJzdWliOiJ1bGlwliwiaWF0IjoxNzI2NjM1NDkwLCJhcH
BzljoiZGF0YXB1c2gifQ.sa9K0laUTG03XiNyegT00w180U2HLn0hWQ60EJd5joO
hs26TynBddx2rkZUEIZBKwG3cPy6gLPx8q2QnU5yXQA' \
--header 'Accept: application/json' \
--header 'Content-Type: application/json' \
--data '{
  "stcode": "",
  "latitude": "14.3473952000",
  "longitude": "78.0001640500",
  "radius": "1"
}'
```

For obtaining an authentication token, refer to [Section 1.4](#)

Response

To acknowledge the above request, ULIP will send the below response -

a) In case of an invalid format

1. Invalid stcode

```
{
  "response": null,
  "error": "true",
  "code": "400",
  "message": "Data format failed OR wrong value entered at: stcode. Format
should follow ^[0-9]{1,6}$"
}
```

2. Invalid latitude and longitude format

```
{
  "response": null,
  "error": "true",
  "code": "400",
  "message": "Data format failed OR wrong value entered at: latitude. Format
should follow ^[0-9]{2}\\.\\{0,1\\}[0-9]{2,10}\\[0-9\\]{0,2}$"
}
```

3. Invalid radius format

```
{
  "response": null,
  "error": "true",
  "code": "400",
  "message": "Data format failed OR wrong value entered at: radius. Format should
follow ^\\d{0,3}(\\.\\d{1,10})?$"
}
```

b) In case MOPNG API data does not exist in the MOPNG system

```
{
  "response": [
    {
      "response": [],
      "responseStatus": "SUCCESS"
    }
  ],
  "error": "false",
  "code": "200",
  "message": "Success"
}
```

c) In case MOPNG API data exist

```
{
  "response": [
    {
      "response": [
        {
          "stcode": 28,
          "diesel": "Diesel",
          "leng": "-",
          "gid": 36288,
```

```

    "villcode": "001962",
    "latitude": 14.347395,
    "dtcode": 502,
    "source": "HPCL",
    "type": "Point",
    "geom": {
      "null": true,
      "type": "geometry",
      "value": "010100000044F312B00280534040899FC8DDB12C40"
    },
    "sdtcode_lg": 5339,
    "sdtcode": 5339,
    "ro_name": "NAGURI FUEL SUPPLIES",
    "auto_lpg": "-",
    "modified": "DD/MM/YYYY",
    "dtcode_lg": 502,
    "state": "Andhra Pradesh",
    "dname": "Sri Satyasai",
    "cng": "-",
    "entity_nam": "HPCL",
    "creation": 44980,
    "longitude": 78.00017,
    "product": "Petrol,Diesel,EV",
    "thumbnail": null,
    "address": "Anantapur,Andhra Pradesh",
    "lng": "-",
    "agency": "HPCL",
    "vilnam_soi": "MUDIGUBBA",
    "address2": "Anantapur",
    "stcode_lg": 28,
    "ro_id": 12678410,
    "sr_no": 30392,
    "cbg": "-",
    "tname": "Mudigubba",
    "ev": "EV",
    "vname": "Taraf Manjali (OG)",
    "active_clo": "Active",
    "petrol": "Petrol",
    "sname": "ANDHRA PRADESH",
    "sublayer_n": null,
    "category": "Diesel",
    "entity": "HPCL",
    "layer_n": null
  },
  {
    "stcode": 28,
    "diesel": "Diesel",

```

```
"lcng": "-",
"gid": 48841,
"villcode": "001962",
"latitude": 14.347395,
"dtcode": 502,
"source": "HPCL",
"type": "Point",
"geom": {
  "null": true,
  "type": "geometry",
  "value": "010100000044F312B00280534040899FC8DDB12C40"
},
"sdtcode_lg": 5339,
"sdtcode": 5339,
"ro_name": "NAGURI FUEL SUPPLIES",
"auto_lpg": "-",
"modified": "DD/MM/YYYY",
"dtcode_lg": 502,
"state": "Andhra Pradesh",
"dname": "Sri Satyasai",
"cng": "-",
"entity_nam": "HPCL",
"creation": 44980,
"longitude": 78.00017,
"product": "Petrol,Diesel,EV",
"thumbnail": null,
"address": "Anantapur,Andhra Pradesh",
"lng": "-",
"agency": "HPCL",
"vilnam_soi": "MUDIGUBBA",
"address2": "Anantapur",
"stcode_lg": 28,
"ro_id": 12678410,
"sr_no": 30392,
"cbg": "-",
"tname": "Mudigubba",
"ev": "EV",
"vname": "Taraf Manjali (OG)",
"active_clo": "Active",
"petrol": "Petrol",
"sname": "ANDHRA PRADESH",
"sublayer_n": null,
"category": "EV",
"entity": "HPCL",
"layer_n": null
},
{
```

```
"stcode": 28,
"diesel": "Diesel",
"lcng": "-",
"gid": 83309,
"villcode": "001962",
"latitude": 14.347395,
"dtcode": 502,
"source": "HPCL",
"type": "Point",
"geom": {
  "null": true,
  "type": "geometry",
  "value": "010100000044F312B00280534040899FC8DDB12C40"
},
"sdtcode_lg": 5339,
"sdtcode": 5339,
"ro_name": "NAGURI FUEL SUPPLIES",
"auto_lpg": "-",
"modified": "DD/MM/YYYY",
"dtcode_lg": 502,
"state": "Andhra Pradesh",
"dname": "Sri Satyasai",
"cng": "-",
"entity_nam": "HPCL",
"creation": 44980,
"longitude": 78.00017,
"product": "Petrol,Diesel,EV",
"thumbnail": null,
"address": "Anantapur,Andhra Pradesh",
"lng": "-",
"agency": "HPCL",
"vilnam_so": "MUDIGUBBA",
"address2": "Anantapur",
"stcode_lg": 28,
"ro_id": 12678410,
"sr_no": 30392,
"cbg": "-",
"tname": "Mudigubba",
"ev": "EV",
"vname": "Taraf Manjali (OG)",
"active_clo": "Active",
"petrol": "Petrol",
"sname": "ANDHRA PRADESH",
"sublayer_n": null,
"category": "Petrol",
"entity": "HPCL",
"layer_n": null
```

```
},
{
  "stcode": 28,
  "diesel": "Diesel",
  "lcng": "-",
  "gid": 177877,
  "villcode": "001962",
  "latitude": 14.347395,
  "dtcode": 502,
  "source": "HPCL",
  "type": "Point",
  "geom": {
    "null": true,
    "type": "geometry",
    "value": "010100000044F312B00280534040899FC8DDB12C40"
  },
  "sdtcode_lg": 5339,
  "sdtcode": 5339,
  "ro_name": "MSHSD NAGURI FUEL SUPPLIES",
  "auto_lpg": "-",
  "modified": null,
  "dtcode_lg": 502,
  "state": "ANDHRA PRADESH",
  "dname": "Anantapur",
  "cng": "-",
  "entity_nam": "HPCL",
  "creation": 0,
  "longitude": 78.00017,
  "product": "Petrol ,Diesel,EV",
  "thumbnail": "DD/MM/YYYY",
  "address": "GANJAPALLI MUDIGUBBA 515511",
  "lng": "-",
  "agency": "HPCL",
  "vilnam_soi": "MUDIGUBBA",
  "address2": "GANJAPALLI MUDIGUBBA 515511",
  "stcode_lg": 28,
  "ro_id": 12678410,
  "sr_no": 766,
  "cbg": "-",
  "tname": "Mudigubba",
  "ev": "EV",
  "vname": "Taraf Manjali (OG)",
  "active_clo": "Active",
  "petrol": "Petrol",
  "sname": "ANDHRA PRADESH",
  "sublayer_n": null,
  "category": "Diesel",
}
```

```
"entity": "HPCL",
"layer_n": null
},
{
  "stcode": 28,
  "diesel": "Diesel",
  "leng": "-",
  "gid": 177939,
  "villcode": "001962",
  "latitude": 14.347395,
  "dtcode": 502,
  "source": "HPCL",
  "type": "Point",
  "geom": {
    "null": true,
    "type": "geometry",
    "value": "010100000044F312B00280534040899FC8DDB12C40"
  },
  "sdtcode_lg": 5339,
  "sdtcode": 5339,
  "ro_name": "MSHSD NAGURI FUEL SUPPLIES",
  "auto_lpg": "-",
  "modified": null,
  "dtcode_lg": 502,
  "state": "ANDHRA PRADESH",
  "dname": "Anantapur",
  "cng": "-",
  "entity_nam": "HPCL",
  "creation": 0,
  "longitude": 78.00017,
  "product": "Petrol ,Diesel,EV",
  "thumbnail": "DD/MM/YYYY",
  "address": "GANJAPALLI MUDIGUBBA 515511",
  "lng": "-",
  "agency": "HPCL",
  "vilnam_soi": "MUDIGUBBA",
  "address2": "GANJAPALLI MUDIGUBBA 515511",
  "stcode_lg": 28,
  "ro_id": 12678410,
  "sr_no": 766,
  "cbg": "-",
  "tname": "Mudigubba",
  "ev": "EV",
  "vname": "Taraf Manjali (OG)",
  "active_clo": "Active",
  "petrol": "Petrol",
  "sname": "ANDHRA PRADESH",
```

```
"sublayer_n": null,
"category": "EV",
"entity": "HPCL",
"layer_n": null
},
{
  "stcode": 28,
  "diesel": "Diesel",
  "lcng": "-",
  "gid": 178001,
  "villcode": "001962",
  "latitude": 14.347395,
  "dtcode": 502,
  "source": "HPCL",
  "type": "Point",
  "geom": {
    "null": true,
    "type": "geometry",
    "value": "010100000044F312B00280534040899FC8DDB12C40"
  },
  "sdtcode_lg": 5339,
  "sdtcode": 5339,
  "ro_name": "MSHSD NAGURI FUEL SUPPLIES",
  "auto_lpg": "-",
  "modified": null,
  "dtcode_lg": 502,
  "state": "ANDHRA PRADESH",
  "dname": "Anantapur",
  "cng": "-",
  "entity_nam": "HPCL",
  "creation": 0,
  "longitude": 78.00017,
  "product": "Petrol ,Diesel,EV",
  "thumbnail": "DD/MM/YYYY",
  "address": "GANJAPALLI MUDIGUBBA 515511",
  "lng": "-",
  "agency": "HPCL",
  "vilnam_soi": "MUDIGUBBA",
  "address2": "GANJAPALLI MUDIGUBBA 515511",
  "stcode_lg": 28,
  "ro_id": 12678410,
  "sr_no": 766,
  "cbg": "-",
  "tname": "Mudigubba",
  "ev": "EV",
  "vname": "Taraf Manjali (OG)",
  "active_clo": "Active",
```

```
{
  "petrol": "Petrol",
  "sname": "ANDHRA PRADESH",
  "sublayer_n": null,
  "category": "Petrol",
  "entity": "HPCL",
  "layer_n": null
},
{
  "responseStatus": "SUCCESS"
}
],
"error": "false",
"code": "200",
"message": "Success"
}
```

Status Code 200: MOPNG API data response

Status Code 400: Bad request (Invalid JSON syntax, invalid JSON key)

Status Code 401 or 403: Unauthenticated or Unauthorized

Status Code 500: Some internal server error occurred

Status Code 502: The server is not responding.

1.3.2 MOPNG Api scheduler

We have implemented a scheduler on the ULIP system that will call the MOPNG API and store the response in the **mopng data dump** database table. This scheduler is configured to execute the API call at midnight on the 1st day of every month.

1.4 Data Transmission & Authentication Mechanism

All data exchanges will be securely done over HTTPS. ULIP's system will request data from MOPNG through a URL with the following format:

For Staging API --

<https://www.ulipstaging.dpiit.gov.in/ulip/v1.0.0/MOPNG/01>

Body:

```
{
  "stcode": "",
  "latitude": "14.3473952000",
  "longitude": "78.0001640500",
  "radius": "1"
}
```

To access the data exchange URL, you would need to use a security mechanism that is implemented by the ULIP system. The process is as follows:

- First time access: you would require a username and password. The username and password would be shared with you beforehand. Once you enter the valid username and password, the ULIP system will return a basic authorization token to you. You can use this token to communicate with the ULIP system until the token gets expired.

In brief, HTTP Basic Authentication works as follows:

When you hit the ULIP secured API for the first time, you need to provide a username and password to get an access token.

```
curl --location
'https://www.ulipstaging.dpiit.gov.in/ulip/v1.0.0/user/login' \
--header 'Accept: application/json' \
--header 'Content-Type: application/json' \
--data-raw '{
    "username": "xxxx",
    "password": "xxxx@123"
}'
```

Authorization:

Bearer

eyJhbGciOiJIUzUxMiJ9.eyJzdWIiOiJ1bGlwliwiaWF0IjoxNzI2NTU2ODk4LCJhcHBzljoiZGF0YXB1c2gifQ.dqXUxA7agKxu9w961sr-QNG1g-9TWWpPQLUX6CsUO5drkwmny9hU4hlf6cOr8CXHIJ67Nv-HdGNtHtr2bH8lrg

After successfully authenticating, the user will be logged into the ULIP system and obtain an access token. This token will be used by the user to communicate with the ULIP system. The token has an expiry time that represents the user session, which is typically set to 30 minutes. If there is no request from the user within this time frame, the session (token) will expire, and the user will need to re-authenticate.

Using The above token, user can request ULIP system for MOPNG API for details as follow -

```
curl --location 'https://www.ulipstaging.dpiit.gov.in/ulip/v1.0.0/MOPNG/01' \
--header 'Authorization: Bearer
eyJhbGciOiJIUzUxMiJ9.eyJzdWIiOiJ1bGlwliwiaWF0IjoxNzI2NTU2ODk4LCJhcHBzljoiZGF0YXB1c2gifQ.dqXUxA7agKxu9w961sr-QNG1g-
9TWWpPQLUX6CsUO5drkwmny9hU4hlf6cOr8CXHIJ67Nv-HdGNtHtr2bH8lrg' \
--header 'Content-Type: application/json' \
--header 'Accept: application/json' \
--data '{
    "stcode": "",
```

```
"latitude": "14.3473952000",  
"longitude": "78.0001640500",  
"radius": "1"  
'
```

For Production API --

<https://www.ulip.dpiit.gov.in/ulip/v1.0.0/MOPNG/01>

Body:

```
{  
  "stcode": "",  
  "latitude": "14.3473952000",  
  "longitude": "78.0001640500",  
  "radius": "1"  
}
```

To access the data exchange URL, you would need to use a security mechanism that is implemented by the ULIP system. The process is as follows:

- First time access: you would require a username and password. The username and password would be shared with you beforehand. Once you enter the valid username and password, the ULIP system will return a basic authorization token to you. You can use this token to communicate with the ULIP system until the token gets expired.

In brief, HTTP Basic Authentication works as follows:

When you hit the ULIP secured API for the first time, you need to provide a username and password to get an access token.

```
curl --location 'https://www.ulip.dpiit.gov.in/ulip/v1.0.0/user/login'  
\br/>--header 'Content-Type: application/json' \  
--header 'Accept: application/json' \  
--data-raw '{  
  "username": "xxxx",  
  "password": "xxxx@123"  
}'
```

Authorization:

Bearer

eyJhbGciOiJIUzUxMiJ9.eyJzdWIiOiJ1bGlwIiwiaWF0IjoxNjM0NzAwLCJhcHBzIjoiaZGF0YXB1c2gifQ.JjmtMU-

**ZFF_zqiifHU8TUp8S9vm5r1H3DbpfXcmIfTU151LLnY_6aAKLMx9p5KJrJxIOdSP0hdNS3D
YHA_2kMg**

After successfully authenticating, the user will be logged into the ULIP system and obtain an access token. This token will be used by the user to communicate with the ULIP system. The token has an expiry time that represents the user session, which is typically set to 30 minutes. If there is no request from the user within this time frame, the session (token) will expire, and the user will need to re-authenticate.

Using The above token, user can request ULIP system for MOPNG API for details as follow -

```
curl --location 'https://www.ulip.dpiit.gov.in/ulip/v1.0.0/MOPNG/01' \  
--header 'Authorization: Bearer  
eyJhbGciOiJIUzUxMiJ9.eyJzdWIiOiJ1bGlwliwiaWF0IjoxNzI2NjM0NzAwLCJhcHBzIjoiaZGF0  
YXB1c2gifQ.JjmtMU-  
ZFF_zqiifHU8TUp8S9vm5r1H3DbpfXcmIfTU151LLnY_6aAKLMx9p5KJrJxIOdSP0hdNS3D  
YHA_2kMg' \  
--header 'Accept: application/json' \  
--header 'Content-Type: application/json' \  
--data '{  
  "stcode": "",  
  "latitude": "14.3473952000",  
  "longitude": "78.0001640500",  
  "radius": "1"  
}'
```